

1 Some Review

Definition 1 A set M with a collection $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ where $U_\alpha \subseteq M$ and $\varphi_\alpha : U_\alpha \rightarrow \mathbb{R}^n$ is called a smooth manifold if the following conditions hold:

- $M = \bigcup_{\alpha_i} U_{\alpha_i}$, where α_i
- φ_α is a bijection on $\varphi_\alpha(U_\alpha)$. The set $\varphi_\alpha(U_\alpha)$ is open in $\mathbb{R}^n$.
- For any $\alpha, \beta \in A$, the transition map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth.
- For any two points $p, q \in M$ there exists two charts (U, φ) and (V, ψ) which are compatible with atlases in the collection, such that $p \in U$ and $q \in V$ and $U \cap V \neq \emptyset$.

Definition 2 We call the following set the n dimensional sphere:

$$S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}$$

2 Stereographic Projection

- I) Consider S^1 . Let $U_N = S^1 \setminus \{(0, 1)\}$ and $U_S = S^1 \setminus \{(0, -1)\}$. Define the map $\varphi_N : U_N \rightarrow \mathbb{R}^n$ as follows. For every point $p \in U_N$, draw a line connecting p to $(0, 1)$. Then $\varphi_N(p)$ is the intersection of that line with the x axis.
- Write down the maps φ_N, φ_S as functions of x, y coordinates of point on circle.
 - What are the images of these maps? Find inverses of these maps.
 - Show that transition maps are smooth.
 - Argue that the rest of conditions for S^1 to be a manifold are satisfied.
- II) Construct a similar stereographic projection for S^2 and show that it gives it a manifold structure as well.
- III) Repeat above, but use polar/spherical coordinates.